Employee Attrition Analysis Using Parametric and Non-Parametric Models

M. Sc. Statistics(Minor Project)

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Abstract

- This study analyzes employee attrition using machine learning techniques.

- Parametric and non-parametric models were implemented and compared.

- Logistic Regression achieved the highest balanced performance.

- Key risk factors influencing attrition were identified.

- Power BI dashboards were developed to support HR decision-making.

Problem Statement

- Employee attrition leads to loss of skilled workforce and high recruitment costs.

- Traditional HR methods rely on surveys and manual assessments.

- These approaches fail to capture hidden patterns in employee data.

- There is a need for a data-driven approach to identify employees at risk of leaving.

Objectives

- Develop predictive models using machine learning
- Compare parametric and non-parametric approaches
- Identify key factors influencing attrition
- Support HR decision-making

Dataset Description

- IBM HR Analytics Employee Attrition Dataset
- Total Records: **1470 employees**
- Total Features: **35 original variables**
- Features include demographic, job-related, and performance attributes
- Target Variable: **Attrition (Yes / No)**
- After encoding categorical variables → 48 features
- Dataset is slightly imbalanced (Attrition rate ≈ 16%)

Sample Dataset(8 Columns)

Age	Attrition	Department	JobRole	JobSatisfaction	MonthlyIncome	OverTime	YearsAtCompany
41	Yes	Sales	Sales Executive	4	5993	Yes	6
49	No	Research & Development	Research Scientist	2	5130	No	10
37	Yes	Research & Development	Laboratory Technician	3	2090	Yes	0
33	No	Research & Development	Research Scientist	3	2909	Yes	8
27	No	Research & Development	Laboratory Technician	2	3468	No	2
32	No	Research & Development	Laboratory Technician	4	3068	No	7
59	No	Research & Development	Laboratory Technician	1	2670	Yes	1
30	No	Research & Development	Laboratory Technician	3	2693	No	1
38	No	Research & Development	Manufacturing Director	3	9526	No	9

- Dataset contains 1470 records and 35 variables.

All 35 Variables

Demographic Factors	Job-Related Factors	Income Factors	Experience Factors
Age	JobRole	MonthlyIncome	TotalWorkingYears
Gender	Department	DailyRate	YearsAtCompany
MaritalStatus	BusinessTravel	HourlyRate	YearsInCurrentRole
Education	JobLevel	MonthlyRate	YearsSinceLastPromotion
EducationField	JobInvolvement	PercentSalaryHike	YearsWithCurrManager
DistanceFromHome	JobSatisfaction		NumCompaniesWorked
	PerformanceRating		TrainingTimesLastYear
	EnvironmentSatisfaction		
	RelationshipSatisfaction		
	WorkLifeBalance		
	OverTime		
	StockOptionLevel		

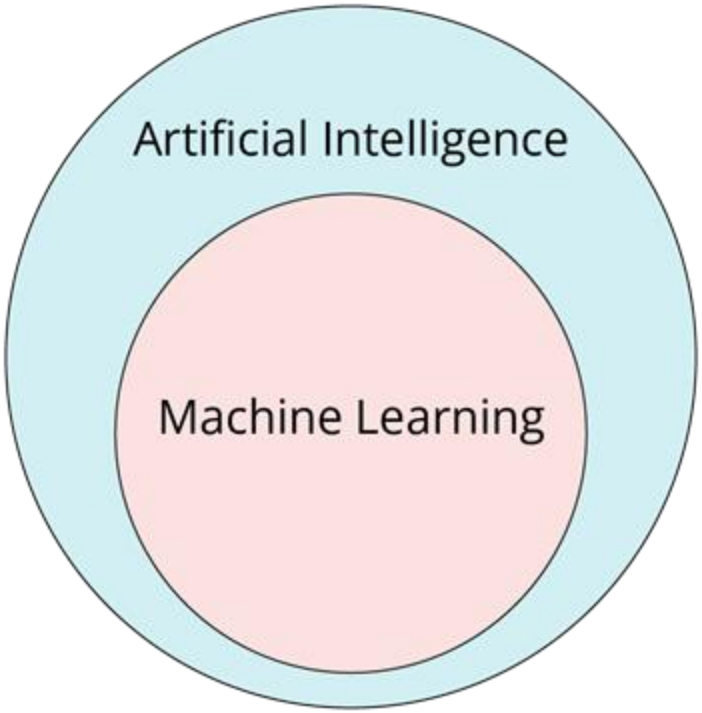
Target Variable: Attrition

Removed During Preprocessing: EmployeeNumber, EmployeeCount, Over18, StandardHours

Descriptive Statistics

	Age	MonthlyIncome	YearsAtCompany	YearsSinceLastPromotion	JobSatisfaction
count	1470.000000	1470.000000	1470.000000	1470.000000	1470.000000
mean	36.923810	6502.931293	7.008163	2.187755	2.728571
std	9.135373	4707.956783	6.126525	3.222430	1.102846
min	18.000000	1009.000000	0.000000	0.000000	1.000000
25%	30.000000	2911.000000	3.000000	0.000000	2.000000
50%	36.000000	4919.000000	5.000000	1.000000	3.000000
75%	43.000000	8379.000000	9.000000	3.000000	4.000000
max	60.000000	19999.000000	40.000000	15.000000	4.000000

Introduction to Machine Learning



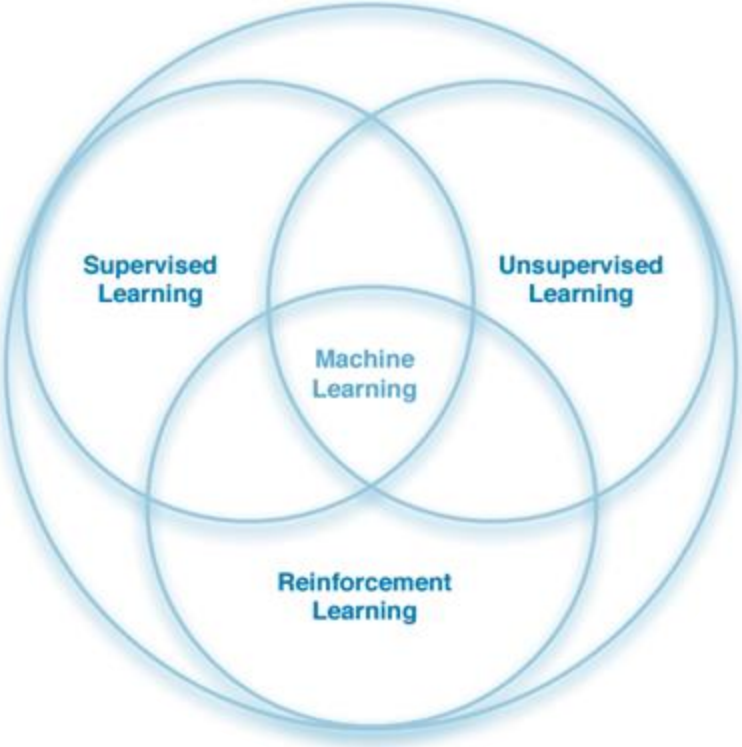
- Artificial Intelligence (AI) is the broader field of creating intelligent systems.
- Machine Learning (ML) is a subset of AI that learns from data.
- ML enables systems to improve performance without explicit programming.
- In this project, we apply Machine Learning to solve an HR problem.

Types of Machine Learning

1) **Supervised Learning** – Uses labeled data to predict outcomes

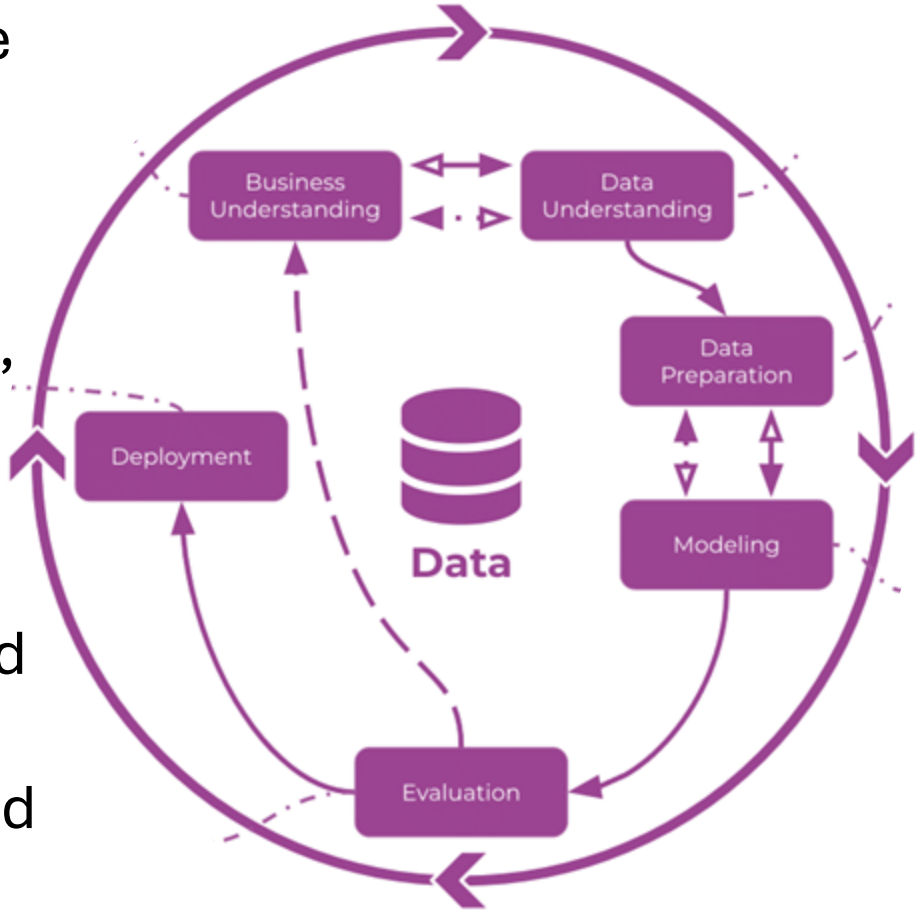
2) **Unsupervised Learning** – Finds hidden patterns in unlabeled data

3) **Reinforcement Learning** – Learns through rewards and penalties



Machine Learning Workflow

- **Business Understanding** – Define attrition prediction problem
- **Data Understanding** – Explore dataset and target distribution
- **Data Preparation** – Encode, scale, and split data
- **Modeling** – Train Logistic Regression, SVM, AdaBoost
- **Evaluation** – Compare models and select best performer
- **Deployment** – Interpret results and support HR decisions



Algorithms Used

Logistic Regression (Parametric Model)

- Statistical model that estimates probability of attrition
- Provides interpretable coefficients to understand feature impact

Support Vector Machine (Non-Parametric Model)

- Constructs optimal decision boundary between classes
- Effective in handling high-dimensional feature spaces

AdaBoost (Ensemble Model)

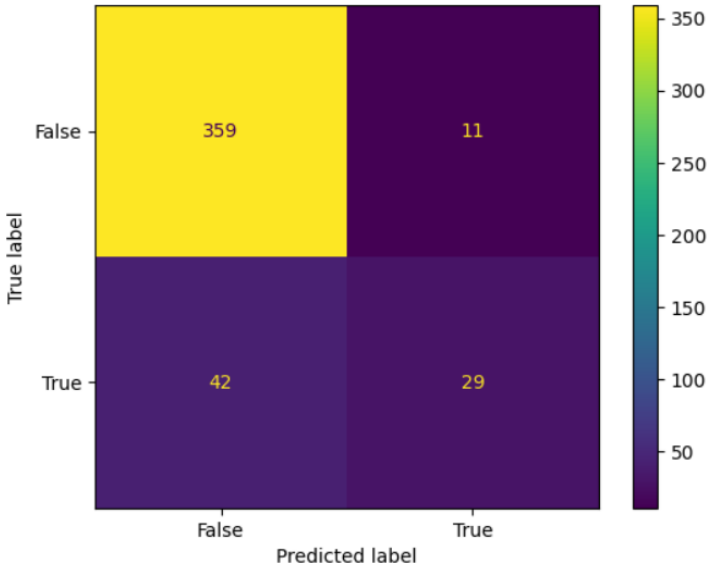
- Combines weak learners to improve prediction accuracy
- Focuses on misclassified observations to enhance performance

Model Performance Comparison

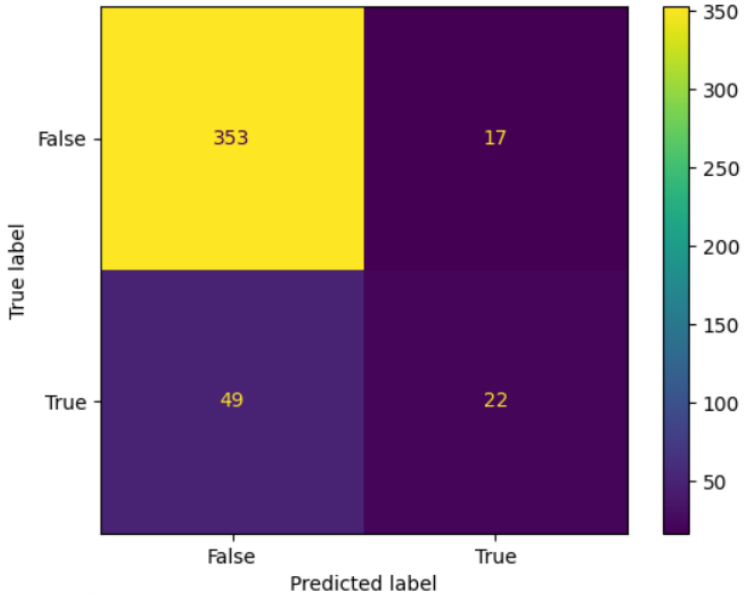
	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.879819	0.725000	0.408451	0.522523
1	AdaBoost	0.850340	0.564103	0.309859	0.400000
2	SVM	0.861678	0.812500	0.183099	0.298851

Logistic Regression achieved the highest accuracy and was selected for further interpretation.

Confusion Matrix Comparison

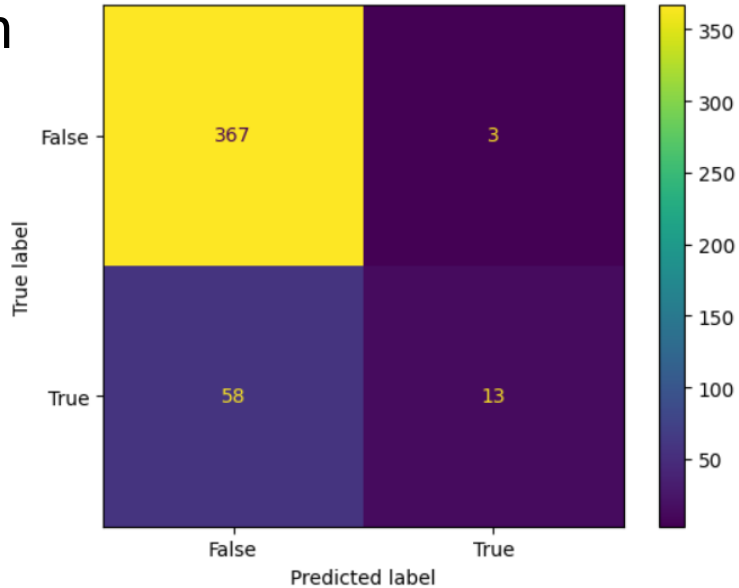


Logistic Regression
(final best model)



AdaBoost

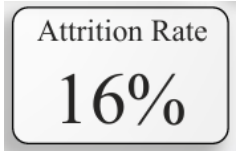
SVM

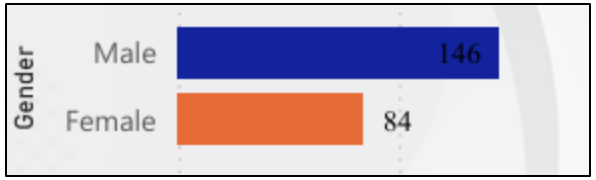


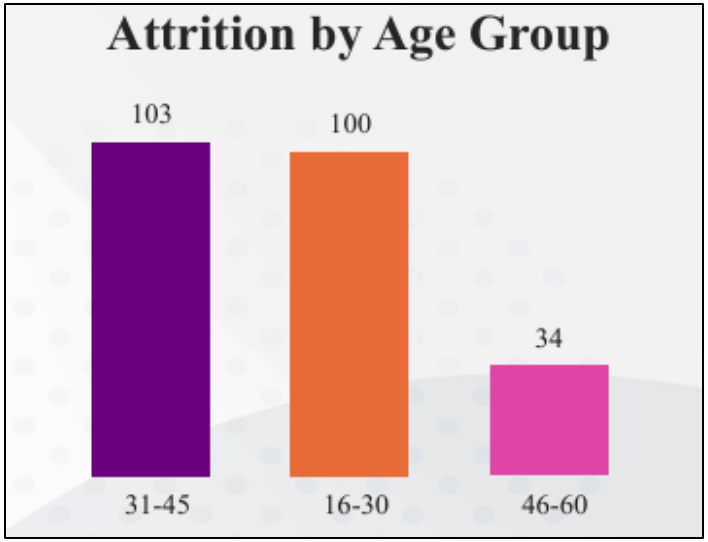
Model Interpretation – Logistic Regression

	Feature	Beta (β)	|Beta|	Effect_on_Attrition	Odds_Ratio
46	OverTime_Yes	0.869160	0.869160	Increases attrition risk	2.384908
26	BusinessTravel_Travel_Frequently	0.690114	0.690114	Increases attrition risk	1.993943
37	JobRole_Laboratory Technician	0.674006	0.674006	Increases attrition risk	1.962081
42	JobRole_Sales Executive	0.619538	0.619538	Increases attrition risk	1.858070
24	YearsSinceLastPromotion	0.616700	0.616700	Increases attrition risk	1.852803
43	JobRole_Sales Representative	0.561369	0.561369	Increases attrition risk	1.753070
13	NumCompaniesWorked	0.492015	0.492015	Increases attrition risk	1.635609
19	TotalWorkingYears	-0.461553	0.461553	Decreases attrition risk	0.630304
6	EnvironmentSatisfaction	-0.451398	0.451398	Decreases attrition risk	0.636737
23	YearsInCurrentRole	-0.437912	0.437912	Decreases attrition risk	0.645382

Executive Summary – Attrition Overview

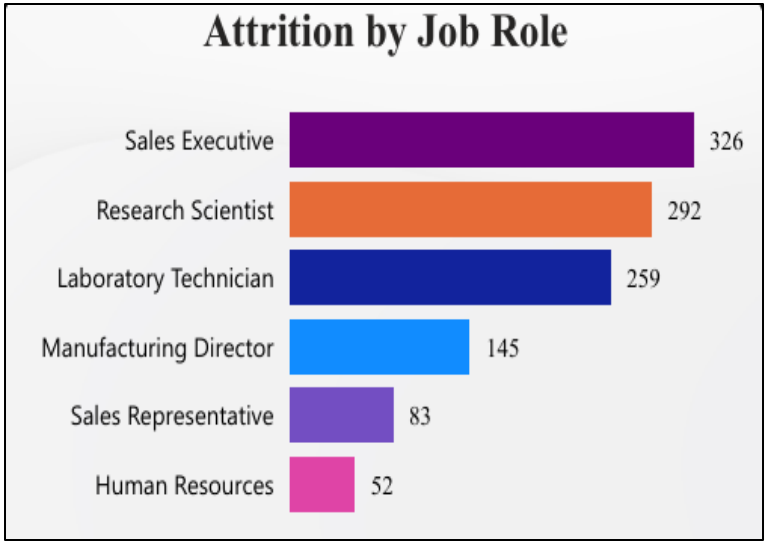






Total employees analyzed: **1470**

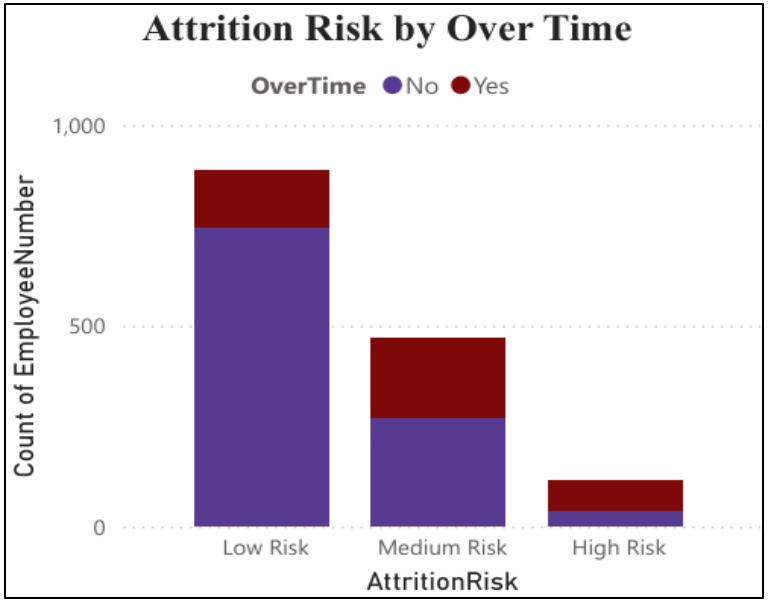
- Overall attrition rate: **16%**
- Higher attrition observed in **Sales and R&D departments**
- Employees aged **31–45 show comparatively higher exits**
- Sales Executive and Research Scientist roles show higher attrition counts

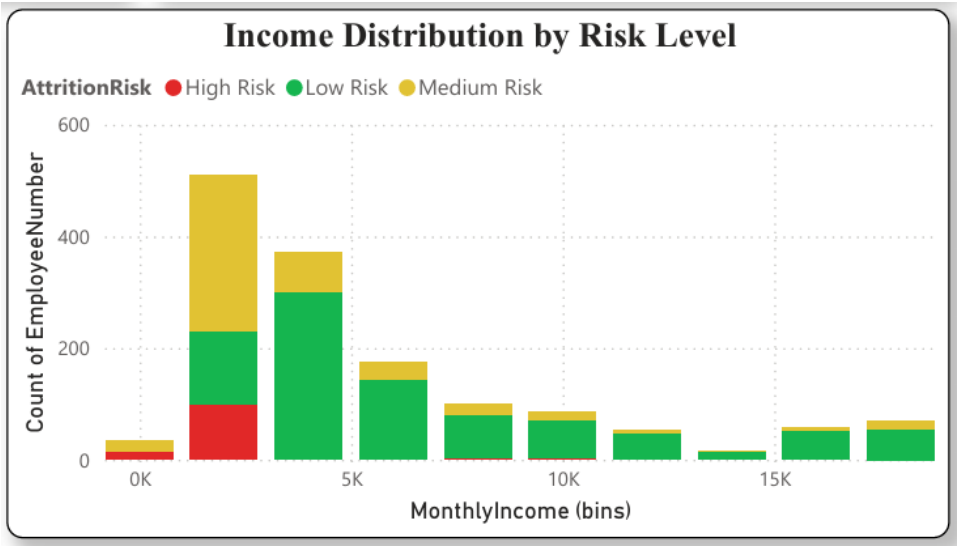


Attrition Risk Drivers

- Employees working overtime show higher concentration in medium and high-risk categories.

- Overtime appears to be a significant contributing factor to attrition risk.





- Higher attrition risk is more prominent in lower income bands.

- Attrition risk decreases as monthly income increases.

Key Business Insights

- Overtime significantly increases employee attrition risk
- Lack of recent promotion contributes to higher turnover
- Employees with lower total experience are more likely to leave
- Certain job roles show consistently higher attrition
- Compensation level influences retention behavior

Recommendations

Reduce excessive overtime workload to improve work-life balance

Implement timely promotion and career growth opportunities

Review compensation structure for lower income groups

Focus retention strategies on high-risk departments and roles

Improve employee engagement and satisfaction initiatives

Limitations & Future Improvements

Limitations

- Dataset is slightly imbalanced (lower attrition cases)
- Model performance evaluated mainly using accuracy and confusion matrix
- Analysis limited to available HR dataset features
- External factors influencing attrition were not included

Future Improvements

- Apply techniques like class balancing to improve minority prediction
- Explore advanced ensemble models for better performance
- Incorporate additional employee engagement variables
- Deploy model as a real-time HR decision support system

Conclusion

Employee attrition was successfully analyzed using machine learning techniques.

Logistic Regression achieved the best performance among the models tested.

Overtime, promotion delay, and compensation were key attrition drivers.

Power BI visualizations supported model findings with business insights.

The model can assist HR in proactive retention strategies.

Thank You